

Gender differences in relation to wayfinding strategies, navigational support design, and wayfinding task difficulty

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ARTICLE INFO

Article history:

Available online 29 July 2008

Keywords:

Navigational support
VR
Weightless
Wayfinding strategy
Gender difference

ABSTRACT

This study investigated user wayfinding navigational performance in terms of two navigational support designs (guide sign and you-are-here map, i.e., YAH), wayfinding strategies (egocentric/route and allocentric/survey), task difficulty (terrestrial/2D or weightless/3D), and gender differences. Eighty participants (40 males and 40 females) were recruited for the wayfinding experiment. The experimental results and statistical analysis indicated that, in the weightless VR scene, navigation time for the terrestrial/2D task type was significantly shorter than for the weightless/3D task type. The guide sign support was significantly more effective than YAH map support. Moreover, the interactions between support style and gender difference indicated that males exhibit better wayfinding performance than females, but that appropriate support can improve navigational performance and eliminate gender differences. In addition, the Way-Finding Strategy Scale result showed that the females were more likely to adopt the egocentric strategy while males were more likely to adopt the allocentric strategy, and that their scores were negatively correlated with navigational performance. Our results can be used to evaluate the interface designs of navigational support systems taking into consideration gender differences with respect to 3D VR games, including VR diving or flying navigational systems.

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1. Introduction

Since the late 1980s, the term “virtual reality” (VR), previously only imagined in science fiction, has signified major progress in computer graphics technology (Rheingold, 1991). The characteristics of VR differ from the real world. For example, constraints such as gravity, solid or linear space isolation, and time are not limited in VR worlds. A VR world allows users to access and jump to different spaces in various ways through the interface provided. Indeed, it extends users’ options by offering the freedom to explore the space according to each user’s needs. Despite the advantage that VR technology offers, previous studies have indicated that, in wayfinding tasks, users do not benefit from the degree of freedom it provides (Elvins, 1997). The major limiting factor of VR wayfinding is that users often get lost or disoriented due to VR space complexity (Darken & Banker, 1998; Darken & Sibert, 1996a, 1996b; Eaton, 1992).

The term “wayfinding” refers to the ability to find a way to a particular destination or location in an expedient manner and to recognize the target when approaching it. Among wayfinding researchers, Lynch (1960) is most credited with developing theories

of real-world wayfinding. From investigations of both real and VR worlds, he identified that cognitive maps of scenes are processed by people during wayfinding (Downs, 1979; Evans & Pezdel, 1980). Cognitive maps indicate the human capacity for recognizing, saving, remembering, and decoding spatial information and location attributes, as well as for forming an action chain of spatial knowledge (Kitchin, 1997).

According to theory relating to cognitive maps, there are two main types of spatial knowledge, namely, route and survey knowledge (Darken & Peterson, 2002; Evans, 1980; Golledge, 1987; Hart & Moore, 1973; Russell & Ward, 1982; Sadeghian, Kantardzic, Lozitskiy, & Sheta, 2006; Siegel & White, 1975). Route knowledge constitutes a sequence of memory about how to get from the starting location to the next place. Survey knowledge is the gestalt-like memory of relationships between locations. Individuals who focus on route knowledge are viewed as dependent on an egocentric wayfinding strategy. Conversely, those focusing on survey knowledge are viewed as applying an allocentric wayfinding strategy (Gramann, Muller, Eick, & Schonebeck, 2005; Thinus-Blanc & Gaunet, 1997).

Klatzky (1998) also identified the use of an egocentric or allocentric reference frame as “a means of representing the locations of entities in space,” leading to distinct spatial representations conveying different types of information (i.e., representational

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primitives). Through three successive experiments, Gramann et al. (2005) further demonstrated that humans tend to prefer either a distinctly egocentric or allocentric reference frame to solve a task by their investigation into two strategic wayfinding groups (turners and nonturners, respectively). Furthermore, they also noted that, when instructed to use the less-preferred reference frame, both groups displayed no decline in response accuracy relative to their preferred frame.

Several studies have demonstrated that various methods can be used for measuring navigation performance and wayfinding strategy around the real and virtual environments (Coluccia & Iosue, 2004; Ruddle & Lessels, 2006). The measuring metrics and methods were varied due to the focus of the researches. As noted that gender differences appear to have a significant effect on wayfinding strategy (Bosco, Longoni, & Vecchi, 2004; Coluccia & Iosue 2004; Cornoldi & Vecchi, 2003; Lawton, 1994, 1996). Dabbs, Chang, Strong, and Milun (1998) suggested that various spatial strategies could interpret and lead to the elimination of gender difference. Using the map drawing test, Coluccia, Iosue, and Brandimonte (2007) argued that males approach maps from a global perspective (allocentric/survey knowledge strategy) while females focus on local features (egocentric/landmarks). This statement was also supported by the literature on the relationship between map drawing and spatial orientation abilities (Coluccia, Bosco, & Brandimonte, 2007).

Additionally, using the Way-Finding Strategy Scale and the Mental Rotations Test, Lawton (1994) argued that males were more likely than females to report using an orientation strategy for wayfinding, whereby one's own position is tracked in relation to geographical reference points. However, females were more likely than males to report using a route strategy, with a focus on the features of a specific route, particularly where a change in direction was required. Among these various navigation methods related to gender difference, some studies have also claimed that time spent on mental processing is key to interpreting the relationships between gender difference, mental maps, and spatial strategies (e.g., Loring-Meier & Halpern, 1999). Thus, in the present study, the search time spent on wayfinding was measured and the Way-Finding Strategy Scale (Lawton, 1994) was used to see how gender and wayfinding strategies affect navigational search time in virtual environments (VE).

In order to improve navigational performance, previous studies have attempted to develop navigational support systems (Bailey, 1994; Darken & Sibert, 1996a, 1996b; Levine, Marchon, & Hanley, 1984; Ruddle, Payne, & Jones, 1998, 1999). Among the related studies, Chen and Stanney (1999) proposed a theoretical model of wayfinding in VR by providing suggestions for navigational support. The model included five categories of navigational supportive tools that can be generalized into two main groups: guide sign and you-are-here (YAH) maps.

This model asserted that guide sign support is more effective than YAH support (Chen & Stanney, 1999; Passini, 1992). In this regard, the authors summarized two main reasons. The first reason is that the YAH is a two-dimensional (2D) oriented support mode; third-directional (3D) information is not provided. Note that most previous studies have focused on two-dimensional (2D) oriented cognitive mapping; little is known regarding the problem of three-dimensional (3D) spatial memory (Ruddle et al., 1999). As Ruddle et al. (1999) have pointed out, the VEs that allow participants to move directly to any position differ from those in which users are constrained to routes delineated by solid barriers such as walls (Vidal, Amorim, & Berthoz, 2004; Vidal & Berthoz, 2005). The issue of task complexity (2D-terrestrial or absent-gravity oriented) related to the comparison of the navigational support mode (YAH or guide sign) remains to be clarified.

The second reason is that the guide sign support mode is egocentric while the YAH is allocentric. A previous study found that

participants use an ego- or allocentric reference frame according to the information they require (either ego- or allocentric) in relation to the VE task (Gramann et al., 2005). As the virtual environment allows participants to move directly to any position, including yaw rotation or third-directional translation, egocentric oriented support provides clearer and more direct supportive information for wayfinders. It would be worth investigating how egocentric (guide signs) and allocentric (YAH) oriented support modes interact with the wayfinding strategy (egocentric or allocentric) while navigating in VEs, including yaw rotations as in terrestrial navigation, and also pitch and roll rotations (i.e., like an astronaut navigating in space).

Additionally, Coluccia and Iosue (2004) pointed out that gender differences appear when the task is more difficult, they disappear when the task is easy. The relationships between navigational support in terms of gender, task difficulty effects and wayfinding strategy also require further clarification.

In the present study, based on the above discussion, we investigated how VR navigational support can influence human navigational performance. Through a wayfinding navigation experiment and self-report questionnaire investigation, the differences between navigational supports (signs and map) in relation to gender, task difficulty (terrestrial/2D and weightless/3D), and wayfinding strategy (route/ego and survey/allocentric strategies) were explored. The following specific results were hypothesized:

- (1) Task difficulty will affect navigation performance.
- (2) Females will be more likely to use the guide signs support mode based on the egocentric (route) wayfinding strategy.
- (3) Males will be more likely to use the YAH map support mode based on the allocentric (survey) wayfinding strategy.
- (4) There will be an interaction between navigational support and task difficulty.
- (5) There will be an interaction between gender wayfinding strategies and task difficulty.
- (6) There will be an interaction between gender wayfinding strategies and navigational support.

2. Materials and design

2.1. Participants

Eighty participants (40 males and 40 females) were recruited for the wayfinding experiment by hierarchical randomization. There were 40 participants with 20 females and 20 males using both of the navigational support conditions, guide signs and YAH map. These participants were college students from the design school of Ming Chuan University, Taiwan, ranging in age from 19 to 26 years ($M = 21.8$, $SD = 2.67$). All participants were paid approximately US\$7 per hour for their participation.

2.2. Navigational supports

A VR exhibition of aquatic life was constructed for the experiment. In the VR world, the exhibition aisles were mainly built of translucent glass, while virtual swimming fish and vivid water effects were simulated (Fig. 1). In the virtual aquarium, 28 objects with exhibition panels were displayed along the translucent aisles at equal distance from one another. However, only four of the exhibited items were assigned for use in the wayfinding tasks.

Three types of navigational settings were designed for the experimental purpose. Type I had no navigational support for learning and practice before the formal test. Type II utilized guide sign support (egocentric reference framework) using 3D arrow signs, which were programmed to point to the assigned target

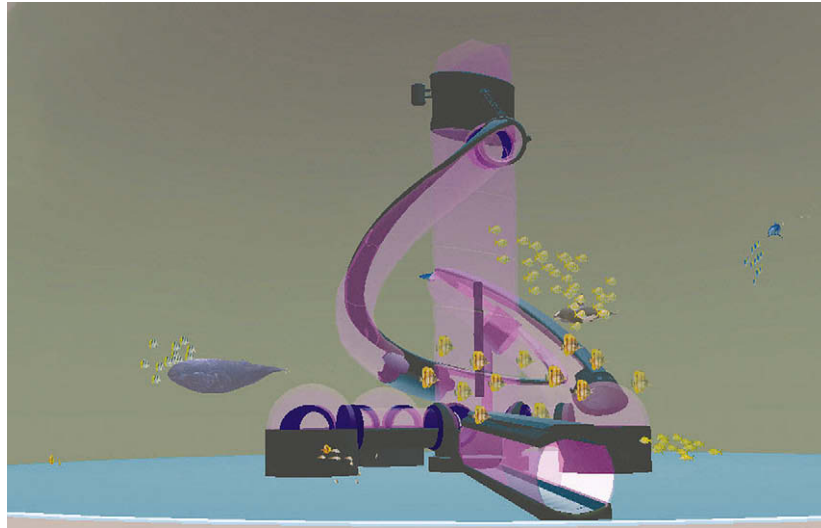


Fig. 1. The VR scene design for experiment.

concurrently with users' movements; users could move toward the target by following the direction of the arrow (Fig. 2a). Type III utilized the YAH map support (allocentric reference framework) shown on the upper left corner of the display, embedded with a square translucent survey map (top view), which indicated the position and orientation of the target (yellow dot) and current user position (red triangle) simultaneously with any movement (see Fig. 2b).

To enable participants to manipulate viewing angle and speed of movement the following settings were programmed: the up and down keystrokes were set as move forward or backward functions, the left and right direction keystrokes were for left and right rotation, and the mouse cursor indicated lens direction and focus.

2.3. Task difficulty

Two tasks corresponding to the task types (terrestrial/2D and weightless/3D) were developed. The specific task types were as follows:

- (1) Terrestrial task: the task was to find a specific object located at the bottom of the VE, with no altitude information (terrestrial/2D-oriented; see Fig. 3, task 2-1).

- (2) Weightless task: to accomplish this task, the participants had to find the object located within the VE, with altitude information (weightless/3D-oriented; see Fig. 3, task 2-2).

2.4. Wayfinding strategies

Based on the tool developed by Lawton (1994), the Way-Finding Strategy Scale was applied to determine whether the participants used the route or the survey strategies. There were 14 questions in the questionnaire. The participants were asked to estimate their typical wayfinding strategies. A 5-point Likert scale (1: strongly disagree and 5: strongly agree) was adopted for measuring the levels.

Factor analysis was conducted for the Way-Finding Strategy Scale, and an initial factor analysis of the scale (principal components, varimax rotation, with two factors specified) was conducted. After excluding the questions with a factor power less than 0.4, nine questions remained; 53.1% of the total variance was explained, the KMO value was 0.71, and the Bartlett test indicated significance (<0.001).

The resulting factor loadings are shown in Table 1. Four items were found to load most heavily on the first factor; these items represented the route (egocentric) strategy (e.g., asking for directions about turning right or left at particular streets or landmarks). Five items loaded most heavily on the second factor; these items represented the survey (allocentric) strategy (e.g., asking for

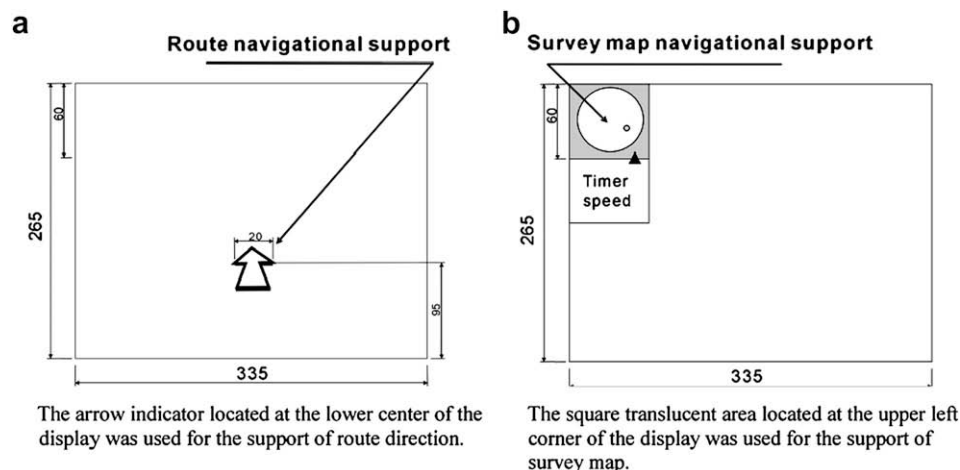


Fig. 2. Two types of navigational supports designed in this study, i.e., (a) guide sign support and (b) you-are-here (YAH) map support.

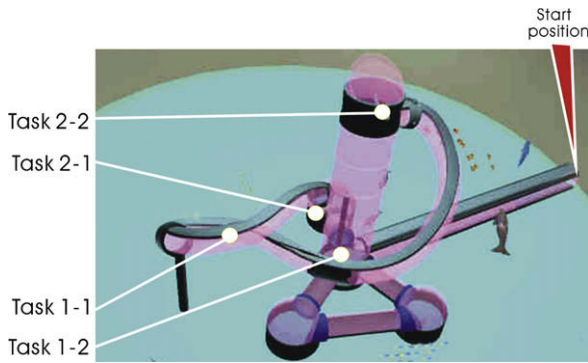


Fig. 3. The experimental scene and locations of the wayfinding task.

holistic image or cognitive map of the wayfinding strategy). This analysis produced a correlation of 0.32 (Kendall's tau-*b* value, $p < 0.01$) between the two factors, showing that the two factors were relatively independent.

The data showed the mean score of route wayfinding strategies for the females to be 4.12 ($SD = 0.512$), which was higher than that for males, at 3.99 ($SD = 0.64$). The mean score of allocentric wayfinding strategies for the females was 2.92 ($SD = 0.677$), which was lower than that for males, at 3.59 ($SD = 0.64$). Data under each factor were then re-coded. Individuals with scores lower than the mean score of the total sample were coded as 0, and those with scores higher than the mean score were coded as 1. The re-coded data were analyzed along with statistics of navigational performance, gender, and navigational support to determine the relationships among them.

2.5. Procedure

This study adopted Virtools (3rd version) software to construct an interactive VR interface. All three navigational settings were run on a Microsoft operating system and were displayed on a 17-in LCD color monitor. The resolution was 1024×768 pixels at a frame rate of 85 Hz. The activities of participants were monitored throughout the experiment. The activities of participants were monitored throughout the experiment.

The specifications of the experiment and task were first briefly explained to participants. Then, they were asked to learn to interact with the type I user interface (no navigational support) by conducting practice tasks. No time constraint was imposed during the warm-up session. The session ended when the participant felt confident in having a basic understanding of the interface.

Table 1
Factor loadings for the Way-Finding Strategy Scale.

Wayfinding strategies	I	II
Route strategy		
Before starting, I asked for directions telling me whether to turn right or left at particular streets or landmarks.	0.79	−0.22
Before starting, I asked for directions telling me how many streets to pass before making each turn.	0.63	−0.07
I made a mental note of landmarks, such as buildings or natural features, that I passed along the way.	0.63	0.42
Before starting, I asked for directions telling me whether to go east, west, north or south at particular streets or landmarks.	0.60	0.19
Survey (orientation) strategy		
I kept track of the relationship between where I was and the center of town.	0.12	0.77
I kept track of the relationship between where I was and the next place where I had to change direction.	0.41	0.75
I visualized a map or layout of the area in my mind as I drove.	0.21	0.73
I kept track of the direction (north, south, east or west) in which I was going.	−0.34	0.61
I kept track of where I was in relationship to the sun (or moon) in the sky as I went.	−0.20	0.55

After the participants had become familiar with the interface, the practice session began. Each task was explained by the research staff. All participants were instructed to complete two of the wayfinding tasks (Fig. 3, task1-1 and task 1-2), i.e., to find and point out the target object within the VR using the interface settings.

At the experimental stage, participants returned two weeks after the practice session. They were randomly divided into two groups with 40 participants each (20 females and 20 males). The participants interacted with only one type of navigational support setting (either guide sign or YAH map). They were all required to participate in another two designated object wayfinding tasks. The location of the designated object was different from that of the practice stage. Two of each of the tasks were given randomly to avoid the carry-over effect; one task was terrestrial/2D-oriented (see Fig. 3, task 2-1), and the other was a weightless/3D-oriented (see Fig. 3, task 2-2) wayfinding task.

Once the participants completed the experiments, they were required to fill out the self-report questionnaire on their psychological perceptions of the wayfinding strategy. The full experiment lasted approximately 1 h for each of the participant.

2.6. Data analysis

The search time in seconds of navigational performance was analyzed using ANOVA. The correlations among navigational supports, wayfinding strategy, and navigational performance were analyzed using regression.

The collected data were analyzed using $2 \times 2 \times 2$ mixed-design ANOVA, with gender (female and male), and 2 navigational supports (guide signs and YAH map) as between-subject factors, and 2 task types (terrestrial/2D or weightless/3D) as within-subject factors. Split data (by between-subject factors) were further analyzed using post hoc paired comparisons to identify the simple interactive effects among the factors. It was observed during the experiment that backtracking frequency and search time varied by task difficulty, with backtracking frequency and search time higher on the weightless/3D-oriented task than for the terrestrial/2D task. Thus, the authors did not analyze backtracking frequency.

Each task was confirmed by the research staff when the task was completed; thus, the authors did not compile the correct rates for analysis. Because no physical constraint was imposed in this VR environment, participants could fly or move across walls or along the vertical axis. With neither correct route nor direction to be followed, the statistics of wrong turns and hesitation frequency were confounded. Previous studies indicated that wayfinding time is significantly correlated with turns and hesitation frequencies (Coluccia & Iosue, 2004). Therefore, neither wrong turns nor hesitation frequencies were analyzed in the present study.

3. Results

3.1. Overall results

Table 2 presents the means and standard deviations of navigation time by split data of task type, navigational support, and gender. The main effect of navigation time for the task type, terrestrial/2D, was

Table 2
Means and standard errors of navigation time.

Gender	Navigational support	N	Task type	
			Terrestrial/2D	Weightless/3D
Female	Guide sign/egocentric	20	48.85 (19.33)	57.95 (19.68)
	YAH map/allocentric	20	155.15 (89.37)	180.75 (100.00)
Male	Guide sign/egocentric	20	65.60 (26.54)	71.55 (37.83)
	YAH map/allocentric	20	84.50 (39.67)	85.55 (42.65)

Unit: second

significantly shorter than for the weightless/3D task ($F(1, 76) = 5.96$, $p < 0.05$). The main effects of between-subject factors, navigational support and gender, were also noted to be significant. Data showed that navigation time for guide signs (egocentric) was significantly shorter than for YAH map (allocentric) support ($F(1, 76) = 32.13$, $p < 0.001$). The main effect of gender on navigation time was significant ($F(1, 76) = 8.59$, $p < 0.001$). The results indicated that males navigated significantly faster than females.

The data revealed a two-way interactive effect, but the three-way interactive effect was not significant in the data collected. The two-way interactive effect was found between gender and navigational support ($F(1, 76) = 18.02$, $p < 0.001$), as shown in Fig. 4. A significant effect was seen on the search time of female participants among the navigational supports. Thus, the females scored their best navigational performance using guide sign (egocentric) support but worst using YAH map (allocentric) support. Nevertheless, no significant difference was found in the search time of male participants for both navigational supports. These data suggest that the type of navigational support significantly affected females, while it did not affect the males' wayfinding navigational performance.

3.2. Correlations

The correlations between navigational performance, wayfinding strategy, and navigational support for the entire sample, and for males and females separately, are shown in Table 3. First, for the female participants, the correlation between navigational performance and navigational support was significant, $r = 0.658$ ($p < 0.001$). The correlation between the two wayfinding strategies (route/egocentric and survey/allocentric) was significant, $r = 0.220$ ($p < 0.05$). A negative correlation was detected between navigational performance and wayfinding strategies. The correlation between navigational performance and route (egocentric) strategy was $r = -0.417$ ($p < 0.05$), and that between navigational performance and survey (allocentric) strategy was $r = -0.276$ ($p < 0.05$). Thus, navigational support modes and wayfinding strategies were all significantly correlated with the females' navigational performance.

Second, for the male participants, a negative correlation was found between navigational performance and survey strategy. The correlation between the two wayfinding strategies (route/egocentric and survey/allocentric) was significant, $r = 0.496$ ($p < 0.01$). Thus, the survey (allocentric) wayfinding strategy was the only spatial knowledge found to be correlated with the males' navigational performance.

3.3. Regression

To further identify a theoretical model among the variables, regression using the enter method was computed. Results indicated

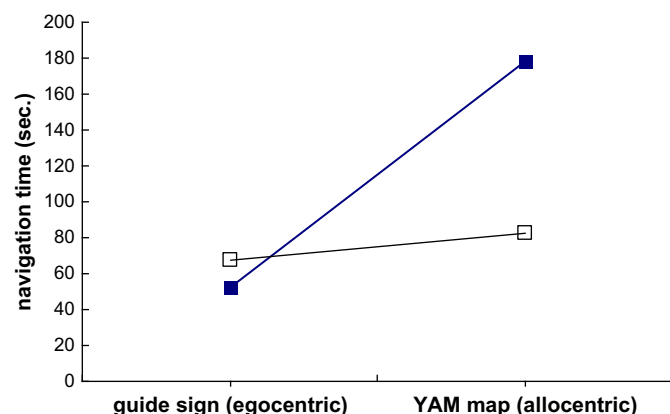


Fig. 4. Effect of sex (■ female, □ male) and navigational supports on navigation time.

the navigational support and wayfinding strategy affected the participants' navigational performance significantly but that the levels would interact with gender differences accordingly. For the females, the theoretical model could explain 78.6% of the variance in navigational performance, $F(3, 36) = 20.79$, $p < 0.01$ (Table 4). As indicated by the individual variable effect, navigational support, route wayfinding strategy, and survey wayfinding strategy all reached statistically significant levels, and the β values were 0.654, -0.332 , and -0.237 ($p < 0.01$). That is, the navigational support affected the females' navigational performance significantly. However, the wayfinding strategies could also affect navigational performance negatively: the higher the score, the worse the navigational support effect on navigational performance.

For males, the theoretical model could explain 50.7% of the variance in navigational performance, $F(3, 36) = 4.16$, $p < 0.05$. As shown by the individual variable effect, only the survey wayfinding strategy reached a statistically significant level, and the β value was -0.546 ($p < 0.01$). That is, the type of navigational support did not affect the males' navigational performance. However, the survey wayfinding strategy could affect the males' navigational performance negatively: the higher the score reported, the poorer the navigational performance.

4. Discussion

4.1. Task complexity and gender difference

Experimental results showed that navigation time for the terrestrial/2D task type was significantly shorter than for the weightless/3D task type. Our first hypothesis, that task difficulty will affect navigation performance, was supported by the research results. This result was also found to be consistent with previous studies (Bosco et al., 2004; Coluccia & Iosue, 2004; Coluccia, Bosco, et al., 2007; Coluccia, Iosue, et al., 2007; Cornoldi & Vecchi, 2003).

However, the interaction between task difficulty and gender, and the interaction between task difficulty and navigational support were both not significant. Thus, the fourth and the fifth hypotheses, that there would be interactions between gender and task difficulty, and between navigational support and difficulty, were not supported by the research results.

One possible reason for this is that the settings of the VE are based on weightlessness (absence of gravity); the participant can travel freely in any direction and can freely rotate. As previous studies have indicated, VEs allowing participants to move weightlessly differ from those that are constrained to terrestrial movement (Ruddle et al., 1999; Vidal et al., 2004; Vidal & Berthoz, 2005). Therefore, the task design for either terrestrial location (located at the bottom of the VE) or weightless location (located with altitude/height within the VE) would all be affected by the weightless VE conditions. Another explanation related to the comparison of the terrestrial and weightless VE conditions is that humans do not innately comprehend weightless conditions (Vidal et al., 2004). Even though the task was located at the bottom of the VE, the participants would not benefit if the VE setting was weightless.

4.2. Gender difference and navigational performance

As the data indicated, the search time using the guide sign support was only half of that needed for YAH map support (guide sign = 64.3 s, YAH map = 133.15 s). Data analysis showed that the guide sign support had a significantly better effect than YAH map support. This phenomenon seems to correspond to the hypotheses of previous studies, which also showed that the guide sign has a superior effect compared to the YAH map (Butler, Acquino, His-song, & Scott, 1993; Levine, 1982; Levine, Jankovic, & Palij, 1982; Passini, 1992).

Table 3

Pearson correlations among gender, navigational support, and Way Finding Strategies (a).

	Gender							
	Female		Male					
	Navigation performance	Navigational support	Route/egocentric strategy	Survey/allocentric strategy	Navigation performance	Navigational support	Route/egocentric strategy	Survey/allocentric strategy
Navigation performance	1				1			
Navigational support	0.66**	1			0.17	1		
Route/egocentric strategy	−0.42*	−0.05	1		−0.07	0.05	1	
Survey/allocentric strategy	−0.28*	0.05	0.22*	1	−0.46**	−0.10	0.50**	1

* $p < 0.05$, ** $p < 0.01$.

Data also indicated that males have better navigational performance than females (females = 119.35, males = 78.55). This phenomenon adheres to the argument elaborated in many studies that males have better spatial capabilities (e.g., Lawton, 1994, 1996; Schiff & Oldak, 1990; Smith & McPhee, 1987).

Additionally, the research also revealed that the interaction between gender and navigational support was significant, and that the type of navigational support only affected the females significantly. Thus, the sixth hypothesis, that there is an interaction between gender wayfinding strategies and navigation support modes, was supported by the present study. However, the gender difference between navigational performances was eliminated when guide sign supports were used, but not with YAH map support. The second hypothesis, that females are more likely to use the egocentric oriented support mode, was supported by our results, whereas the third hypothesis, that males are more likely to use the allocentric oriented support mode, was not shown. A possible explanation and correlation among the factors will be discussed below.

4.3. Correlations among the variables

As the regression analysis further indicated, the correlations among gender, navigational support, and wayfinding strategy were significant. We noted that navigational support affected the females' navigational performance significantly, but both route and survey wayfinding strategies had a significantly negative correlation with navigational performance. The higher the wayfinding strategy scores the females reported, the poorer their navigational performance was found to be. A possible explanation for this phenomenon is that the higher wayfinding scores indicated that participants possessed better comprehension of specific wayfinding strategies, which could have confounded the effect of the navigational support systems. As only guide sign navigational support affected the females' navigational performance, it appears that

females tended to be egocentric (route) strategy wayfinders. For the males, only the survey wayfinding strategy was negatively correlated with navigational performance. The higher the survey wayfinding strategy scores the males reported, the poorer their navigational performance. This indicates that males tended to be allocentric strategy wayfinders. This result is consistent with a previous study that suggested that females are egocentric strategy wayfinders, and males are allocentric wayfinders (Lawton, 1994).

Nevertheless, no significant navigational support effect was found on the males' navigational performance. Although the Way-Finding Strategy Scale indicated that males are allocentric wayfinders, the YAH map support did not significantly affect the males' navigation performance. As noted previously, a possible explanation is that YAH support may be impaired if the VE is weightless-oriented (Vidal et al., 2004; Vidal & Berthoz, 2005) and, thus, males may not benefit from the YAH support. Another possible explanation is that instead of relying on the navigational support systems (guide signs or YAH map support), male participants tended to complete the wayfinding task using their own spatial capacity. Previous studies have also argued that the two wayfinding strategies may not be totally independent (Lawton, 1996; Gramann et al., 2005), and that people who possess relatively good wayfinding capacities may shift between wayfinding strategies during navigation (Kyllonen, Lohman, & Woltz, 1984).

Note that this experiment did not include the case of using the two support types simultaneously. In addition, a previous study has also pointed out that coexisting navigational support systems might obscure their comparison (Gramann et al., 2005). Thus, we could not demonstrate that the participants paid attention to only one of the two simultaneous support systems offered in the present study.

A future enhancement of the present study would be to include other kinds of navigational interface systems (complexity of the interface control) (Lessels & Ruddle, 2005; Wraga, Creem-Regehr, & Proffitt, 2004), measuring metrics (measurement of user behavior or neural signal) (Creem-Regehr, Neil, & Yeh, 2007; Ruddle & Lessels, 2006) and other VE features (field of view/FOV or fidelity of VE) (Creem-Regehr, Willemssen, Gooch, & Thompson, 2005; Lessels & Ruddle, 2005) to further pinpoint the interactions among VE characteristics, gender, navigational support mode, and wayfinding strategies.

5. Conclusions

The experimental results and statistical analysis indicate that in the weightless VR scene, task difficulty did affect the wayfinding performance significantly. Navigation time for the terrestrial/2D task type was significantly shorter than for the weightless/3D task type. The guide sign support was significantly more effective than the YAH map support. Moreover, the interactions between support

Table 4

Coefficients (a).

	Female		Male	
	Standardized coefficients		Standardized coefficients	
	β	t	β	t
Navigational support	0.654	6.46**	0.110	0.76
Route/egocentric strategy	−0.332	−3.20**	0.192	1.16
Survey/allocentric strategy	−0.237	−2.29*	−0.546	−3.27**
R^2		0.796		0.507
F		20.79		4.16
df		3, 36		3, 36
p		0.001		0.013

* $p < 0.05$, ** $p < 0.01$.

style and gender difference indicate that males exhibit better wayfinding performance than females, but that an appropriate support system can improve navigational performance and eliminate gender differences. In addition, the Way-Finding Strategy Scale result showed that the females were more likely to adopt the egocentric/route strategy and the males were more likely to adopt the allocentric/survey strategy, and that their scores were negatively correlated with navigational performance. Our results can be used to evaluate the interface designs of navigational support systems, taking into consideration gender differences on 3D VR games, and including VR diving or flying navigational systems.

Acknowledgments

Financial support of this research study by National Science Council under the grant NSC 95-2221-E-011-046 and NSC 95-2218-E-011-008 are gratefully acknowledged.

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